

OPEN SOURCE · CANADA (US EXPERIMENTAL) · RUNS ON YOUR MACHINE

# taxjson

Your broker's CSV in. Schedule 3 out. Capital gains, dividends and the superficial-loss rule computed the way the CRA defines them — across every account you own — with every rule citing its source.

```
bash -c "$(curl -fsSL https://taxjson.com/install.sh)"
```

Installs `taxjson` and its short name `tjs` · [taxjson.com](https://taxjson.com) · [github.com/taxjson/taxjson](https://github.com/taxjson/taxjson) · MIT

## THE PROBLEM

# The slip your broker files is not your cost basis

A T5008 reports proceeds per account. The Income Tax Act taxes you on **one average cost per security across all your accounts**, denies losses when any affiliated account — your RRSP included — bought the same thing within 30 days, and folds option premiums, splits, spin-offs and return of capital into that cost. Nobody's statement does that math for you.

### Cross-account pooling

s.47: one ACB pool per identical property. A US listing and its TSX listing are the same pool; a CDR is not.

### Superficial losses

s.54 / s.40(2)(g): a loss is denied if you *or an affiliated person* bought within  $\pm 30$  days and still holds at day 30 — sized sale by sale. Denied into an RRSP, it is gone for good.

### Everything that moves cost

Assignments (IT-479R), warrant exercises, splits, renames, spin-offs (s. 86.1), mergers with elections, ROC, reinvested distributions — in the right order, on the settlement date.

**Spreadsheets get the easy 90% right. The other 10% is where reassessments come from.**

## WHAT IT DOES

# Three steps, all on disk

### inputs/

Broker CSV exports per account: IBKR, Questrade, RBC, Webull, Kraken, Coinbase — plus plain-text `.tt` for anything else



### tjs run

Parse · convert at the Bank of Canada rate per settlement date · pool identical securities · dated renames and corporate actions with elections · superficial-loss pass across taxable and registered accounts



### reports/

Per-account summaries, Schedule 3 / Form 8949 rows, the wash radar, T1135, a tax and minimum-tax estimate — and the broker cross-check

### Two project files

`taxjson.toml` : year, country, accounts (taxable / sheltered) and the figures you enter by hand. `ticker.map` : dated renames and symbol lookups.  
`tjs migrate` converts an older project.

### Row accounting

Every parser accounts for every row it reads. Rows it recognises but does not translate are counted and named — never silently dropped.

### Traceable numbers

`taxjson audit` recomputes every disposition from the parsed broker row through FX, ACB and the wash determination, and ties it out against the reports.

# Every computation names its source

| RULE                        | WHAT TAXJSON DOES                                                                                                                                                                                                                                                         | SOURCE                                            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Identical properties</b> | One average-cost pool per security across accounts; cross-listings merged, CDRs kept apart; a ticker change carries the pool on its date                                                                                                                                  | s.47(1); IT-387R2; T4037                          |
| <b>Superficial loss</b>     | CRA's formula, sale by sale: the least of the units sold, bought in the 61-day window, and held at day 30; the RRSP/TFSA is affiliated; denied-into-plan is permanent                                                                                                     | s.54, s.40(2)(g)(i), s.53(1)(f), s.251.1(1)(g)    |
| <b>Timing</b>               | Sales on the settlement date, including at year end; a dividend in the year paid, a Canadian trust's distribution in the year payable                                                                                                                                     | T4037; s.82(1); s.104(13)                         |
| <b>Foreign currency</b>     | Bank of Canada rate on each settlement date, never an annual average                                                                                                                                                                                                      | Folio S5-F4-C1                                    |
| <b>Options</b>              | A written option's premium is a gain in the year written; a buy-back is a loss in its own year; an assignment folds the premium into the shares; an exercised warrant's cost goes into the shares too. <code>option-boundary</code> says when a filed year needs a T1-ADJ | s.49(1)–(4); IT-479R                              |
| <b>Corporate actions</b>    | Splits, spin-offs, mergers with the rollover election recorded and a filing reminder at every run; ROC reduces ACB                                                                                                                                                        | s.86.1, s.85.1, s.53(2)(h)                        |
| <b>US (experimental)</b>    | FIFO; §1091 wash sales with basis and holding-period carry, IRA repurchases permanent; lots carried between your own accounts; §355 / §356 per lot; Form 8949 codes                                                                                                       | §1091, Rev. Rul. 2008-5; Reg. 1.358-2, 1.356-1(b) |

`tjs tax-logic` prints every rule taxjson applies, one line each, with the id its tests cite — Canada's and the USA's apart, never mixed. REFERENCES.md maps each to its source, including what is deliberately *not* modelled. A rule that disagrees with its source is a bug; that is the most useful issue the project can receive.

## WORKED EXAMPLE

# A loss you cannot claim

### What happened

The margin account sells 400 shares at a loss on February 3. On February 10 the RRSP buys 200 of the same stock, and still holds them on March 5.

### What the slip says

A capital loss on 400 shares. The spreadsheet agrees.

### What the Act says

CRA's formula: the least of 400 sold, 200 acquired in the 61-day window and 200 still held at its end — the loss on 200 shares is superficial (s.54). The ACB bump goes to the RRSP, where it is worthless: permanently denied. Shares the RRSP held before the window would not have counted.

**NOTE:** superficial loss – 200 sh denied (permanently: replacement property held in rrsp)  
loss sale 2026-02-03 · trigger lot 2026-02-10 (rrsp) · window end 2026-03-05

**Only a tool that sees every account at once can get this right. That is the whole design.**

# Checked against the one thing that cannot lie

Internal reports can all agree and still be wrong. `taxjson sanity` compares the positions the books say you hold with the positions your broker's own export says you hold, account by account. Any difference is a trade that has not reached the CSV yet, or a booking problem — there is no third kind.

## 6,500+ tests

Every fix lands with a regression test that failed before it, tagged with the tax-logic rule it pins; the gate fails on a rule with no test.

## Fuzzers and mutants

Three property fuzzers run thousands of random ledgers per run; mutation testing of both engines finds the boundaries no test pins.

## Two full audits

About 1,300 and then 1,600 findings, after eight review rounds and a security audit. Every confirmed finding reproduced, fixed, pinned.

## Published limits

KNOWN\_ISSUES.md lists what is out of scope and what to do by hand, with the citation.

## COMMANDS

# One pipeline, a few dozen views on it

### Set up

- `init` a project
- `migrate` an old one
- `fetch` (plugin)
- `elect` elections

### Build the books

- `run` the pipeline
- `crypto-sends`
- `find-missing-history`

### Summaries

- `sum` by form line
- `estimate` / `amt`
- `stats` / `instalments`
- `fx-cash`

### Positions

- `list` per account
- `shares` all accounts

### Row listings

- `events` / `trades`
- `divs` / `dil` / `roc`
- `gains` / `fees` / `leaps`
- `transfers`

### Totals by type

- `winners` ranked
- `divs-sum` / `dil-sum`
- `trades-sum` / `fees-sum`
- `ccd-sum` / `leaps-sum`
- `roc-sum`

### Before you trade

- `wash-radar` windows
- `buy-check`
- `sell-check`
- `harvest` / `scan`
- `watch` for cron

### Before you file

- `checklist` every step
- `form-export` / `t1135`
- `reconcile-slips`
- `carryover` / `handoff`
- `close-year`
- `check-filed`

### Explain and check

- `audit` / `tax-logic`
- `renames` / `splits`
- `spinoffs` / `wash-sales`
- `sanity` / `check-dates`
- `edge-cases`

### Tools

- `redact` an export
- `help` (`--all`)

The groups are the help page's: `tjs` with no command lists every command under them, one line each — inside a project only your country's. Every command takes `-h` ; most take `--json` . `tjs redact` strips identifiers from a real export so it can be shared as a parser sample.

# Nothing personal leaves the machine

## Local by construction

Parsing, pooling and the wash pass run on your computer — no server, no web UI, just commands and files. The pipeline's only network calls are FX rates (Bank of Canada, its noon rate before March 2017; Yahoo only before May 2007) and crypto prices for unpriced rows — both switchable off with `TAXJSON_OFFLINE=1`. SECURITY.md lists every egress.

## No credentials in the core

The core holds no broker client and reads no credential. Broker fetch is a separate, optional package (taxjson-fetch): it uses your own token file, never writes it into a project, and sends it nowhere but that broker.

## The project guards itself

A personal-data scan runs in every gate and as the pre-push hook, fails closed, and reads a private denylist kept outside the repository; the hook also refuses a commit message that quotes a money total. `redact` keeps every row shape and drops the identifiers it recognises — review before sharing.

**Zero runtime dependencies in the core. A tax tool has to be trustworthy before it is clever.**



## SCOPE

# Canada is the product. The US engine is a draft.

### Canada — supported

Average-cost ACB, the superficial-loss rule across accounts, settlement-date timing, options and warrants, dated renames, corporate actions with elections, T1135, Schedule 3, a marginal-rate estimate with provincial tables, minimum tax line by line, instalments, and carry-forwards from one year's lock to the next. Validated against two years of real multi-account books and the brokers' positions.

### US — experimental

FIFO, §1091 wash sales across your accounts, lots carried between your own accounts, §355 / §356 per lot, short/long-term, Form 8949 codes and a TXF export. Implemented and unit-tested, never validated on a real account. It says so at every run. Real US exports — redacted — are the way to change that.

The two rule sets never mix: a project has one country, and a setting, flag or command the other country owns is refused, not silently applied.

**Honest labelling costs nothing. A tax tool that overclaims loses the only thing it has.**

BROKERS

# Reads the exports you already have

| BROKER               | FORMAT                                            | NOTES                                                                                                                     |
|----------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Interactive Brokers  | Activity / Flex statement CSV                     | Trades, options, corporate actions, warrant exercises, transfers, fees, dividends; Flex auto-fetch (taxjson-fetch plugin) |
| Questrade            | Activity CSV                                      | Trades, DRIP, cash-in-lieu, transfers; REST auto-fetch and live positions (taxjson-fetch plugin)                          |
| RBC Direct Investing | Activity export                                   | Trades, dividends with withholding, warrant exercises                                                                     |
| Webull               | Trading Summary CSV                               | Trades and options (BUY/SELL rows only); no income — T5 interest and dividends go in as <code>.tt</code> lines            |
| Kraken · Coinbase    | Ledgers / transaction history                     | Trades, rewards, sends and withdrawals as custody evidence                                                                |
| Anything else        | <code>.tt</code> plain text · generic CSV mapping | Ten-line files a human can write and read                                                                                 |

Help wanted: Wealthsimple, TD Direct, BMO InvestorLine, CIBC Investor's Edge, Scotia iTRADE, National Bank Direct, Qtrade, Desjardins. A parser is an afternoon's work given one real export — `tjs` `redact` makes sharing one safe. An auto-fetcher for another broker is a plugin of its own.

## INSTALL

# Your first books in ten minutes

```
bash -c "$(curl -fsSL https://taxjson.com/install.sh)"
```

Installs the latest release into its own Python environment and links `taxjson` and its short name `tjs` onto your PATH. Needs git and Python 3.9+, nothing else. Re-run to upgrade; add `--with-fetch` for the Questrade / IBKR auto-fetch plugin.

```
mkdir -p ~/taxes/2026 && cd ~/taxes/2026
tjs init --country canada      # taxjson.toml, ticker.map, inputs/
# drop broker CSVs into inputs/margin/, inputs/rrsp/, ...
tjs run                        # books, reports, wash pass, broker check
tjs sum                        # the year, by Schedule 3 line
tjs wash-radar                 # "can I sell this at a loss today?"
tjs checklist                  # at year end: ready to file?
```

WHERE IT GOES

# Roadmap

## More Canadian brokers

Driven by redacted real exports from users. One parser per afternoon, each with the row-accounting contract; auto-fetch for more brokers as plugins.

## Rules retail investors hit

Non-eligible dividends and T3 box splits, the s.39(4) election, CNIL — each stated in tax-logic with its tests, whichever real exports show first.

## US validation

The engine exists; it needs a real account. The label comes off when the numbers have been checked against one.

## 1.0

PyPI, a frozen CLI, and the interchange contract with portfolio tools written down.

[taxjson.com](https://taxjson.com) · [github.com/taxjson/taxjson](https://github.com/taxjson/taxjson) · MIT · not tax advice — a calculator that shows its work.